```
print("hello, machine learning!")
print ("iam learinin python using google colab.")
```

```
hello, machine learning!
iam learinin python using google colab.
```

```
name = "zahra"
age = 27
height = 160
is_student = True

print (name)
print (age)
print (height)
print(is_student)

print(type(name))
print(type(age))
print(type(height))
print(type(is_student))
```

```
zahra
27
160
True
<class 'str'>
<class 'int'>
<class 'int'>
<class 'bool'>
```

```
a = 7
b = 2

print("Addition:", a + b)
print("Subtraction:", a - b)
print("Multiplication:", a * b)
print("Division:", a / b)
print("Floor division:", a // b)
print("Remainder:", a % b)
print("Power:", a ** b)
```

```
Addition: 9
Subtraction: 5
Multiplication: 14
Division: 3.5
Floor division: 3
Remainder: 1
Power: 49
```

```
name = input("enter your name:")
age = int(input("enter your age: "))

print("hello", name)
print("next year you will be", age +1)
```

```
enter your name:zahra
enter your age: 27
hello zahra
next year you will be 28
```

```
mark = float(input("Enter your mark: "))

if mark >= 60:
  print("Pass")
else:
  print("Fail")
```

```
Enter your mark: 70
Pass
```

```
marks = [75, 82, 91, 68, 88]

print("Marks:", marks)
print("First mark:", marks[0])
```

```
print("Third mark:", marks[2])
print("Number of marks:", len(marks))
print("Total:", sum(marks))
print("Highest:", max(marks))
print("Lowest:", min(marks))
print("Average:", sum(marks) / len(marks))
```

```
Marks: [75, 82, 91, 68, 88]
First mark: 75
Third mark: 91
Number of marks: 5
Total: 404
Highest: 91
Lowest: 68
Average: 80.8
```

```
temperature = [25.4, 26.1, 27.3, 28.0, 26.8]

print("Number of observations:", len(temperature))
print("Average temperature:", sum(temperature) / len(temperature))
print("Minimum temperature:", min(temperature))
print("Maximum temperature:", max(temperature))
```

```
Number of observations: 5
Average temperature: 26.72
Minimum temperature: 25.4
Maximum temperature: 28.0
```

```
study_hours= [1,2,3,4,5,6,7]
marks = [50,55,60,65,72,80,88]

for i in range (len(study_hours)):
  print("study hours:", study_hours[i], "marks:", marks[i])
```

```
study hours: 1 marks: 50
study hours: 2 marks: 55
study hours: 3 marks: 60
study hours: 4 marks: 65
study hours: 5 marks: 72
study hours: 6 marks: 80
study hours: 7 marks: 88
```

```
marks = [50, 55, 60, 65, 72, 80, 88]

for mark in marks:
  if mark >= 60:
     print(mark, "Pass")
else:
  print(mark, "Fail")
```

```
60 Pass
65 Pass
72 Pass
80 Pass
88 Pass
88 Fail
```

```
study_hours = [1, 2, 3, 4, 5, 6, 7]
marks = [50, 55, 60, 65, 72, 80, 88]

num_samples = len(marks)
print ("Number of samples: " + str(num_samples))

avg_hours = sum(study_hours) / float(num_samples)
print ("Average study hours: %.2f" % avg_hours)

avg_marks = sum(marks) / float(num_samples)
print ("Average mark: {:.2f}".format(avg_marks))


highest_mark = max(marks)
lowest_mark = min(marks)
print ("Highest mark: " + str(highest_mark))
print ("Lowest mark: " + str(lowest_mark))

for i in range(num_samples):
    if marks[i] >= 60:
        status = "Pass"
```

```
    else:
        status = "Fail"
    print ("Student " + str(i+1) + ": Hours = " + str(study_hours[i]) + ", Mark = " + str(marks[i]) + " -> [" + status + "]")

passes = 0
for mark in marks:
    if mark >= 60:
        passes += 1

print ("\nNumber of students who passed (60 or above): " + str(passes))
```

```
Number of samples: 7
Average study hours: 4.00
Average mark: 67.14
Highest mark: 88
Lowest mark: 50
Student 1: Hours = 1, Mark = 50 -> [Fail]
Student 2: Hours = 2, Mark = 55 -> [Fail]
Student 3: Hours = 3, Mark = 60 -> [Pass]
Student 4: Hours = 4, Mark = 65 -> [Pass]
Student 5: Hours = 5, Mark = 72 -> [Pass]
Student 6: Hours = 6, Mark = 80 -> [Pass]
Student 7: Hours = 7, Mark = 88 -> [Pass]

Number of students who passed (60 or above): 5
```

1. How would you modify the program to work with 10 or 20 samples?
- since we know the number of samples we can esally update the loop limit or sit the size.
2. How would you count the number of marks below 60?
- by using the conditional statment inside the loop.
3. How would you find the percentage of students who passed?
- by dividing the count of passing marks over the total number of marks then multiply it by 100.
4. How would you add another feature, such as attendance percentage?
- by adding a new list of percentages under attendance percentage name.
5. If study hours are the input feature and marks are the target, what type of relationship might you expect between them?
- positive relation, if you study more you will get higher marks and vice versa.
6. Why is it useful to store data in a structured form before applying a Machine Learning algorithm?
- data need to be organized so the machine could understand.